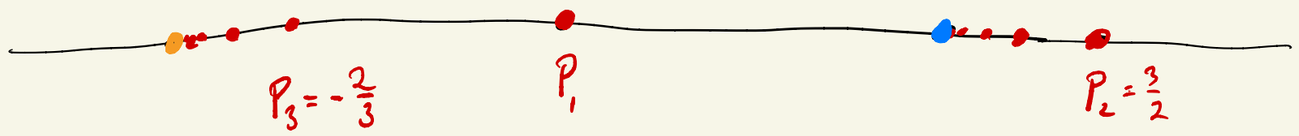
Topics

- Subsequences
- Subsequences + Compactness
- Cauchy Criterion + completeness

Def A <u>subsequence</u> of a sequence (p_n) is a sequence (p_{n_i}) where $n_1 < n_2 < n_3 < \cdots$. If $\lim_{i\to\infty} p_{n_i} = s$, s is a <u>subsequential limit</u> of (p_n).

Example $p_n = (-1)^n + \frac{1}{n}$

$0, \left(1+\frac{1}{2}\right), \left(-1+\frac{1}{3}\right), \left(1+\frac{1}{4}\right), \left(-1+\frac{1}{5}\right), \ldots$



Two convergent subsequences:

$s_n = p_{2n-1} = 1 + \frac{1}{2n-1} \longrightarrow 1$ $\qquad n_i = 2i-1$

$t_n = p_{2n} = -1 + \frac{1}{2n} \longrightarrow -1$ $\qquad n_i = 2i$

Thm (3.6)

(a) Every sequence in a cpct. metric space has a convergent subsequence.

(b) Every bounded seq in $\mathbb{R}^d$ has a convergent subseq.

proof: (a) Let (p_n) be a sequence in a compact metric space K.

Suppose finite image $\{x_1, \dots, x_k\}$. p_n must equal x_j for infinitely many n, for some $j \in \{1, \dots k\}$, so (p_n) has a constant subsequence.

Suppose infinite image. Image is infinite subset of cpct $\Rightarrow$ has a limit pt. p.

Having chosen $n_1 < n_2 < \cdots < n_{i-1}$, choose $n_i > n_{i-1}$ s.t. $d(p_{n_i}, p) < \frac{1}{i}$.

$$\frac{1}{i} \to 0 \quad \Rightarrow \quad d(p_{n_i}, p) \to 0 \quad \Rightarrow \quad p_{n_i} \to p$$

(b) Bounded in $\mathbb{R}^d$ $\Rightarrow$ contained in compact set (e.g. closed ball).

Let E be the set of subsequential limits of (p_n).

What can E look like?

- Can $E = [0,1]$?
- Can $E = \mathbb{R}$?

Thm (3.7) The subsequential limits of a sequence (p_n) form a closed subset.

proof: Let $E = \{\text{subsequential limits of } (p_n)\}$.

Let q be a limit point of E.



Suppose we have chosen $n_1 < n_2 < \cdots < n_{i-1}$ s.t. $d(p_{n_j}, q) < \frac{1}{j}$ for $j = 1, \ldots, i-1$.

Choose $s \in E$ s.t. $d(s,q) < \frac{1}{2i}$.

There are infinite $n \in \mathbb{N}$ s.t. $d(p_n, s) < \frac{1}{2i}$. Let n_i be one of these greater than n_{i-1}.

We have constructed $n_1 < n_2 < \cdots$ s.t. $d(p_{n_i}, q) < d(p_{n_i}, s) + \frac{1}{i} \quad \frac{1}{i}$

So $p_{n_i} \longrightarrow q$.

When can we expect a seq to converge?

In "nice" metric spaces, "diminishing jumps" gives convergence.

Def (p_n) in X is a **Cauchy sequence** if $\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$ such that $d(p_n, p_m) < \varepsilon$ for all $m, n > N$.

Def The **diameter** of $E \subset X$ is

$$\operatorname{diam} E = \sup \{ d(p,q) \mid p, q \in E \}$$

Lemma
(p_n) is cauchy iff diameter of its tails goes to 0.

$$\operatorname{diam} \{ p_N, p_{N+1}, \ldots \} \longrightarrow 0$$

Thm (3.11a) Every convergent seq. is Cauchy.

proof: Suppose $p_n \longrightarrow p$.

Let $\varepsilon > 0$.

$\exists N$ s.t. $n \geq N \Rightarrow d(p_n, p) < \frac{\varepsilon}{2}$.

So $m, n \geq N$ implies

$$d(p_m, p_n) \leq d(p_m, p) + d(p, p_n) < \varepsilon$$



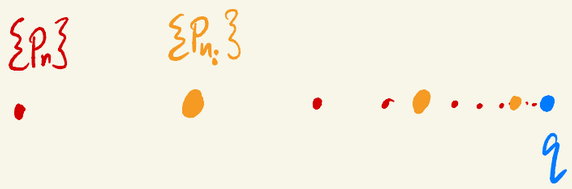
Do Cauchy sequences always converge?

Not in every metric space!

- $p_n = \frac{1}{n}$ does not converge in $(0, 2)$
- $p_n = \left(1 + \frac{1}{n}\right)^n \longrightarrow e$, so doesn't converge in $\mathbb{Q}$.

Thm If q is a subsequential limit of a cauchy seq (p_n) then $p_n \to q$.

proof: Suppose $n_1 > n_2 > \cdots$ and $p_{n_i} \to q$.



Let $\varepsilon > 0$.

$\exists N$ s.t. $m,n > N \Rightarrow d(p_m, p_n) < \frac{\varepsilon}{2}$ because (p_n) is cauchy.

$\exists i$ s.t. $n_i > N$ and $d(p_{n_i}, q) < \frac{\varepsilon}{2}$ because $p_{n_i} \to q$.

Suppose $n \geq N$.

$$d(p_n, p) \leq d(p_n, p_{n_i}) + d(p_{n_i}, q) < \varepsilon$$

Def A metric space X is <u>complete</u> if every cauchy sequence in X converges.

Examples

- $\mathbb{Q}$ is not complete
- $(0,2)$ is not complete

Next time

- Compact metric spaces are complete
- $\mathbb{R}^d$ is complete